Research Method of Data Science

Iegor Vyshnevskyi, Ph.D.

Sogang University

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Your individual final research project

Week 10 progress conversation

What you should have now, what we will build next, and how the paper comes together.

Our destination: **a small, coherent study that you can explain and support with evidence.**

Agenda

OUR PROJECT CONVERSATION

1 · Destination
What the final project should accomplish

2 · Today
Where your project should be in Week 10

3 · Paper
What belongs in each section

4 · Path
How Weeks 11–14 build the draft

5 · Finish
Presentation, revision, and submission

Appendix
Checklists, project map, and obstacles

1. What I expect from the final project

> Your project investigates **one focused question** using evidence you can actually obtain, understand, and explain.

- ✓ A manageable scope and a clear reason for the study
- ✓ Relevant literature connected to the question
- ✓ A transparent approach to evidence and analysis
- ✓ Findings that answer the question within stated limits
- ✓ A complete, readable paper with meaningful content

A complete structure with a manageable scope

Keep the scope realistic

I do **not** expect a full-scale, publication-ready journal article.

Limit the setting, period, dataset, cases, or documents.

Keep the study complete

Every required section must still do its job.

A heading plus “I will analyse this later” is not a finished section.

Depth comes from explaining your decisions and evidence clearly—not from making the project large.

One question connects the whole paper

Literature	→	Evidence	→	Method	→	Results	→	Meaning
What do we know?		What can I observe?		How will I answer?		What did I find?		How far can it go?

Alignment check: if these parts address different questions, the paper needs revision.

2. Where you should be in Week 10

1	2	3	4
Question	**Review**	**Evidence**	**Workflow**
Working and scoped	Organised by themes	Candidate source documented	Preparation and initial inspec

The wording can improve. The direction should already be clear.

What a usable question and review look like

Research question

The reader can identify:

what you examine · **for whom/what** · **where** · **when**

Literature review

The reader sees:

themes · agreements or disagreements · relevant methods · why you question follows

Your review should already contain connected writing and verified citations. Continue updating it as design develops.

What we are finalising in the data work

Quantitative study

Source, coverage, unit and variables
Meanings, units and missingness
Preparation, joins, exclusions and checks
What the sample can represent

Qualitative study

Evidence-selection logic
Context and source provenance
Ethical handling
Transparent analysis trail

You can already start the Data and Method sections

Write Data now

Describe the evidence you selected and the preparation decisions you made.

Begin Method now

Explain the design and the analysis your question requires.

As we learn and apply the next methods, revise the explanation to match what you actually do.

Keep unresolved decisions visible in your working draft.

3. The final paper structure

1 · Introduction
Problem, question, approach

2 · Literature
Themes and framework

3 · Evidence
Source, scope, preparation

4 · Method
Procedure and justification

5 · Results
Findings that answer the question

6 · Discussion
Meaning and alternatives

7 · Conclusion
Direct answer and contribution

8 · Limitations
Boundary of the answer

9 · References
Verified sources

Add the required AI-use disclosure and reproducible supporting materials.

Introduction and literature review: why the study follows

Introduction · orient

- Define the problem and why it matters.
- State the specific research question.
- Preview the approach.
- Once known, state the main answer and contribution.

Literature · position

- Organise scholarship through themes and connections.
- Show what is established or debated.
- Explain how concepts and methods inform the design.
- Define the limited issue your study examines.

Start both now. Finalise the introduction after analysis, and use original, verified sources throughout.

Data and method: what you studied and what you did

Data / evidence · document

- Source, selection, coverage, and period
- Variables or materials and definitions
- Preparation, missingness, and exclusions
- Access, ethical, and representativeness limits

Method · justify

- Describe the steps connecting evidence to the answer.
- Define comparisons, variables, assumptions, or coding.
- Explain why the procedure fits the question.
- Report what you actually did—not every available method.

A small evidence overview can help. Choose methods because they fit the question—not because software offers them.

Results: show the evidence and explain it

1 · Orient
State what the reader is seeing

→

2 · Identify
Name the relevant pattern or result

→

3 · Interpret
Explain its substantive meaning

Use actual output. A plot or software screenshot without explanation leaves the result unfinished.

Discussion and conclusion have different jobs

Discussion · interpret

Connect findings to the question and literature.

Examine explanations and alternatives.

Consider implications in proportion to the evidence.

Conclusion · answer

Give the direct, concise answer.

State the contribution and main boundary.

End with the key takeaway or justified next step.

Both sections should follow from the results already presented.

Limitations: explain the boundary of the answer

Name	→	Explain	→	Respond
What is the limitation?		What consequence does it have?		How did you handle it?

Example:

> “The records describe regional averages. They cannot establish whether the same relationship holds for individual residents.”

Other limits can involve selection, measurement, missingness, time coverage, interpretation or causa identification.

A small project can still form a complete study

ILLUSTRATIVE DESIGN

Question	→	Literature	→	Data + method	→	Meaning
How did internet use change in Korea and Vietnam, 2000–2023?		Digital access and its measurement		Annual indicators, trends, endpoint comparisons		Contextualise patterns; keep claims descriptive

4. How the next weeks build your paper

Week 10	Week 11	Week 12	Week 13	Week 14
PREPARE Data section and initial outputs	COMPARE Method choices and first results	ANALYSE Appropriate analysis and interpretation	CHECK Refined method and claim boundaries	CONNECT One complete, coherent draft

Use the approaches relevant to your question.

From working sections to the Week 14 composite draft

Now
Question · review · evidence · method

→

After analysis
Results · discussion · limitations

→

Week 14
Connect every section into one paper

- ✓ Question, literature, evidence, and method align
- ✓ Results answer the question
- ✓ Discussion and conclusion follow from results
- ✓ Citations, definitions, and supporting files agree

5. Presentation, final submission, and what matters

The **35% final examination assignment** combines your individual paper, reproducible supporting materials, and complete-study presentation.

Weeks 15–16

Present the complete research argument and answer immediate questions.

Use feedback to revise before the Week 16 submission.

What matters

Clear question · feasible scope · relevant literature · fitting evidence and method · accurate interpretation · honest limitations

Remember: a simple, well-justified study can demonstrate all of these qualities.

Your next steps and our progress conversation

For your current project:

1. State the working question and point to your review draft.
2. Explain what evidence is ready and what remains unresolved.
3. Identify the next analysis decision and the section you can write now.

Bring specific obstacles to class or office hours: **Tue/Thu 15:00–16:30; Wed 10:00–12:00**, or by appointm

Appendix: a section-by-section completion check

Use the following slides while drafting and revising.

They explain what a reader should learn from each section and what supporting materials should ma
visible.

You do not need to make every section equally long. Give each section enough space to do its job.

The working paper checklist

Introduction
Problem · question · motivation · approach

Literature
Themes · verified citations · rationale

Data / evidence
Source · selection · definitions · preparation

Method
Procedure · justification · current status

Results
Checked findings · explained outputs

Discussion
Literature · interpretation · alternatives

Conclusion
Direct answer · contribution

Limitations
Specific, consequential boundaries

Supporting work
References · disclosure · reproducibility

By the final submission, replace plans with an account of completed work.

A project map for checking alignment

Question	→	Literature	→	Evidence	→	Method	→	Answer
My question is ...		helps me understand ...		covers ...		can show ...		is limited by ...

Before analysis, leave the result unresolved rather than inventing an answer.

Common obstacles and feasible responses

Topic too broad

Narrow the population, period, setting, or relationship.

Data do not fit

Find a justified measure—or revise the question.

Method too advanced

Use a simpler approach that still answers a useful question.

Results unexpected

Verify the workflow and report the evidence honestly.

Review reads like a list? Reorganise it around themes that inform your design.

Discuss major changes early enough to keep the project feasible.

Supporting materials and reproducibility

Provide a clear starting point and a way to follow your work.

Include, as relevant:

- source log, variable or coding definitions, and permissions notes
- preparation and analysis notebooks/scripts or a qualitative audit trail
- the figures, tables or deidentified extracts used in the paper
- a short README explaining files and how to reproduce outputs

Preserve raw evidence where sharing and licensing permit.

Writing tools and references

Use **MS Word or Overleaf**, whichever helps you write and revise clearly.

Use **Zotero or another reference manager** to organise citations and references.

Check that:

- every cited source appears in the reference list
- quotations, paraphrases, data and borrowed figures have attribution
- tables and figures have clear labels and are discussed in the text
- the final formatting follows the posted instructions

Responsibility and the final Week 14 check

AI responsibility

- Disclose the help used and how you checked it.
- Verify sources, code, measures, and every reported finding.
- Be ready to explain the study in presentation and Q&A.

Draft responsibility

- Follow one question from beginning to end.
- Give every section meaningful content.
- Trace findings to evidence and analysis.
- Make introduction and conclusion match the completed study.

Bring the connected draft and supporting materials. Fabricated references, data, or results are unacceptable.